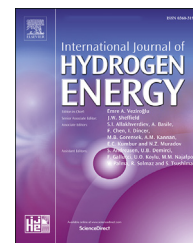


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Decarbonizing Singapore via local production of H₂ from natural gas

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HIGHLIGHTS

- A Multi-Period hydrogen supply chain optimization model is developed.
- Decarbonizing Singapore via local production of blue H₂ is investigated.
- The local production capacity could reach as high as 32,000 TPD to meet all H₂ demand.
- For marine transport, LNH₃ is less expensive than LH₂.
- H₂ fuel cell electric vehicles should be considered for heavy-duty vehicles.

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ABSTRACT

Interest in low emissions hydrogen as an energy vector to assist in deep decarbonization goals has gained momentum recently. In this paper, we explore local hydrogen production from natural gas with CO₂ capture and sequestration (known as “blue” H₂) to support Singapore's intended inclusion of low-carbon intensity H₂ fuel as a way to achieve significant CO₂ emission reduction through 2050. A superstructure-based model is used to minimize the total cost or emissions of the entire supply chain, from H₂ production to consumption in the power, industry, domestic, marine, and road transport sectors. H₂ demand for each sector is projected for three H₂ penetration scenarios (low, medium, and high). The results are analyzed for the levelized cost of H₂, reduction in carbon emissions, and cost of carbon avoidance. Costs associated with direct and indirect CO₂ emissions, as well as H₂ and CO₂ infrastructure costs, are included in total costs. A comparison of blue H₂ and direct natural gas utilization with post-combustion CO₂ capture and sequestration is also presented. We find that decarbonizing Singapore via local production of blue H₂ will involve a relatively high carbon avoidance cost.

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Introduction

The concept of a hydrogen economy, where hydrogen is a carrier for energy storage and transportation [1], [2] and a fuel for the power, heat, and mobility [3] sectors, has been proposed for decades. However, the current use of hydrogen is mainly limited to feedstock for ammonia production [1] and refining [4,5]. But as the world seeks low-carbon technologies to meet CO₂ emissions targets set by the Paris Agreement [6], hydrogen is expected to play a significant role [1,7,8]. Hydrogen roadmaps have been proposed by many countries (United Kingdom [9], United States [10], Australia [11], Japan, etc.) to chart the transition to a sustainable hydrogen economy. These roadmaps propose a hydrogen supply chain (HSC) comprising the infrastructure for production, transportation, storage, and distribution that would support a hydrogen economy in each country. A robust and cost-effective HSC is a prerequisite for the success of a H₂ economy. Its planning involves a series of decisions, aimed at answering the questions of when, where, what sizes and what technology to deploy [12]. These decisions are intertwined with many combinations. An optimization-based tool can help decision makers to rank potential alternatives.

The literature on modeling and optimizing an HSC is extensive. Comprehensive reviews can be found in Agnolucci and McDowall [13] and Li et al. [14]. To the best of our knowledge, the first work on HSC design integrating production, storage, and distribution in a single framework was initiated by Almansoori and Shah [15]. They established a steady state “snap-shot” model to determine a future HSC for the transport sector in Great Britain. Their model suggested that an optimal HSC might consist of centralized steam methane reforming (SMR) plants with hydrogen delivered as a liquid via tanker trucks. Extension of this work by other researchers [14] includes implementing multi-period optimization [16–18], multi-objective optimization [17–19], and introducing uncertainty [19,20]. Almansoori and Shah [16] extended their model to consider the variation of hydrogen demand over a long-term planning horizon, and the availability of raw materials and their logistics. They further extended their model for the design of an HSC under demand uncertainty by a scenario-based approach [20]. De-León Almaraz et al. [17] developed a multi-period optimization framework for hydrogen deployment in the Mid-Pyrenees region in France. Later, the model was tested on a national geographic scale [18]. Three criteria—cost, environmental impact, and safety risk—were optimized by the x1d700-constraint method. The ArcGIS spatial tool was adopted to identify geospatial issues. Similarly, Kim and Moon [19] defined a total relative risk of a hydrogen infrastructure and provided useful guidelines for designing an HSC with improved cost efficiency and safety. Kim et al. [21] proposed a stochastic formulation based on a two-stage programming approach to examine the total cost of an HSC with uncertain hydrogen demand. Guillén-Gosálbez et al. [22] developed a bi-criterion model, which addressed simultaneous optimization of both cost and environmental impact. A bi-level algorithm was also proposed to expedite the search of Pareto solutions. Moreno-Benito et al. [23] studied the development of a sustainable hydrogen infrastructure for the

transport sector in the United Kingdom. The problem was solved by an extension of SHIPMod [24] (Spatial Hydrogen Infrastructure Planning method), which consists of a multi-period spatially-explicit mixed integer linear programming formulation. The results identified hydrogen production from natural gas through SMR with carbon capture as the most cost-effective option to maintain low carbon emissions. Parker et al. [25] evaluated the economic potential and infrastructure requirements for a green energy pathway, where hydrogen was produced from agriculture residues. They found that hydrogen produced via a gasification process from agricultural wastes with co-production of a small amount of electricity can be cost competitive with hydrogen from natural gas, using northern California case studies. Johnson and Ogden [26] developed the Hydrogen Production and Transmission (HyPAT) model, and optimized the interconnected regional pipeline networks between production facilities and demand locations.

Recently, Ogumerem et al. [27] compared two scenarios, where oxygen co-produced with hydrogen from water electrolysis was discarded or sold as a by-product. Yáñez et al. [28] investigated the contribution of upcycling surplus hydrogen from the chlor-alkali industry and steel mills to design an HSC in Northern Spain. Bique et al. [29] developed a multi-criterion optimization model, where costs, safety, and environmental impacts are balanced. A three-dimensional Pareto front was created using the ϵ constraint method. The result of the case study indicated that decentralized hydrogen production by water electrolysis may compete with a coal-based dominant technology. Reuß et al. [30] introduced a methodology for designing an HSC with spatial resolution, which covers various hydrogen transportation and seasonal long-term storage [31]. They investigated hydrogen infrastructure for 2050 in Germany through a comparison of three scenarios, but the evolution of hydrogen infrastructure to keep up with the increasing hydrogen market was not considered. Li et al. [32] studied hydrogen economy in the transportation sector of Franche-Comté, a region in eastern France, covering the entire HSC from feedstock supply to hydrogen refueling stations. Seo et al. [33] compared centralized and decentralized storage modes for hydrogen supply chains in Korea. The role of central storage was predicted to become important as the market share of fuel cell vehicles reaches 15–30%. The work by both Li et al. [32] and Seo et al. [33] comprised a single-period HSC design study, where the evolution of hydrogen demand and infrastructure were not considered. A national supply chain generally focuses on a few H₂ production technologies subject to the local energy resource availability, while a regional supply chain can capture the benefits of integrating energy-abundant areas with high energy-consuming areas in order to overcome any local energy resource limitation. Kawasaki Heavy Industries, Ltd. (KHI) proposed a CO₂-free H₂ supply chain, where hydrogen produced in Australia by coal gasification with carbon capture and storage will be delivered to Japan as liquid hydrogen [34]. Heuser et al. [35] also carried out a techno-economic analysis of a potential CO₂-free hydrogen supply chain between Patagonia and Japan via liquid hydrogen.

Most previous work has focused on country-specific or area-specific supply chains, such as for the UK [15], Germany

[30], France [36], or South Korea [33], and considered hydrogen demand in the transport sector only, neglecting that hydrogen can also be utilized in power and industry sectors. In addition, technology-focused studies have elaborated the competitiveness of H_2 production technologies using different evaluation criteria, such as H_2 cost and CO_2 emissions. So far, there is no literature evaluating all these different technologies on an evolving H_2 supply chain with multiple H_2 demand sectors.

In light of the aforementioned concerns, we study the evolution and development needs of a hydrogen economy in Singapore through 2050. Singapore, as a city-state with only 700 square kilometers of land, relies on imported energy. Imported natural gas generates more than 95% of its electricity. As Singapore is exploring the potential of hydrogen to reduce her carbon footprint, importing it and producing it locally via existing natural gas facilities are two available alternatives. In this study, we explore the possibility of producing hydrogen in Singapore to deep decarbonize transport, power, industry, and domestic sectors. The variation of hydrogen demand over a long-term planning horizon and cost reductions arising from technological advances are considered. Several what-if scenarios are explored to derive insights on relative emissions and costs associated with various hydrogen pathways in Singapore. For convenience, a linear carbon tax scenario is adopted to introduce a penalty for CO_2 emissions, but this is only one example of a policy mechanism that could be adopted to discourage CO_2 emissions.

Hydrogen supply chain in Singapore

Among the available H_2 production routes from fossil fuels, steam methane reforming (SMR) with CCS is considered for local H_2 production in Singapore, using the natural gas (PNG) supplied by the existing natural gas transmission pipelines and liquefied natural gas (LNG) imported. The structure of a typical H_2 supply chain is depicted in Fig. 1. Four entities are considered: H_2 production, H_2 terminal storage, H_2 utilization, and CO_2 sequestration site. H_2 is piped from SMRCCS plants to the terminals, where H_2 is compressed to 40 bar

(CH40) for pipeline distribution to power plants, the industry gas network, and the domestic gas network. H_2 is also converted to liquid ammonia (LNH_3) for utilization as a marine fuel. Captured CO_2 is delivered to a specified CO_2 subsurface formation for sequestration. All CO_2 emissions from the entire supply chain are considered, and penalty is paid for carbon emissions at estimated carbon tax rates. Methane emissions of natural gas supply chain are also considered, which are quantified by CO_2 equivalent emissions when calculating emissions of hydrogen produced by natural gas. Details can be found in Tables 1–1 of the Supporting Information.

Based on the H_2 supply chain structure of Fig. 1, the model is formulated as a mixed-integer non-linear programming (MINLP) problem, minimizing the total cost (TC) and/or the total CO_2 emissions (TCO₂) of the entire H_2 supply chain. A mixed-integer linear programming (MILP) solution of the original MINLP problem is also obtained by linearizing the non-linear terms for the facility costing equations, since MILP models are generally easier to solve. The cost of H_2 , associated emissions, and required infrastructure deployment are identified for an ‘ H_2 economy’ in Singapore through 2050. Detailed techno-economic data of production and terminal entities can be found in Section An of the Supporting Information, while detailed problem statement and model formulations can be found in Section D of the Supporting Information.

Projection for Singapore's H_2 demand through 2050

Based on the data from 2017, we conclude that CO_2 emissions in Singapore [37] are dominated by three major sectors: transport (13%), industry (46%), and power (39%). We focus on using H_2 to decarbonize these sectors and for the heating needs in the industry and buildings. Future H_2 demand is projected based on the current energy demand and projected H_2 penetration, as will be explained in subsequent sections. The adoption of H_2 is assumed to follow an S-shaped logistic substitution model [13,38], as shown in Eq. (1),

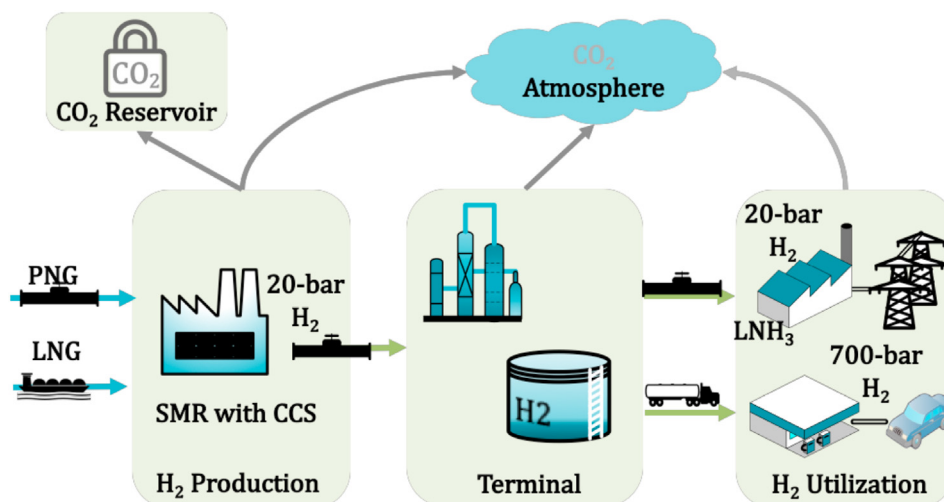


Fig. 1 – Structure of a typical H_2 supply chain.

Table 1 – Vehicle population by type of fuel used in Singapore, 2020.

Type	Gasoline	Diesel	Gasoline-Electric	Diesel-Electric	Electric
Car ^a	572,132	18,076	42,599	18	1217
Taxi	21	6508	9117	–	32
Bus ^b	14	18,798	–	50	50
Light-duty truck ^c	5816	91,085	1	7	97
Medium-duty truck ^c	–	23,176	–	–	–
Heavy-duty truck ^c	–	20,601	–	–	–
Motorcycle	140,782	–	–	–	1

^a Gasoline-Electric (plug-in) and Gasoline-CNG are included in Gasoline- Electric, since the number is negligible.

^b More than 99% of buses use diesel. The public bus population is approximately 5100.

^c Trucks are classified into three types based on their maximum laden weight. Currently, only 97 electric trucks exist in Singapore. For simplicity, we assume that all medium-duty and heavy-duty trucks are diesel-types.

$$N_t = \frac{K}{1 + e^{-\alpha t - \beta}} \quad (1)$$

where N_t and K represent the market penetration in year t and final market penetration (limit of growth) respectively; α specifies the steepness of the S-shape curve; and β specifies the time to reach 50% K . In a practical application, the logistic equation, N_t is also defined as:

$$N_t = \frac{K}{1 + e^{\left[-\frac{\ln(81)}{\Delta t}\right](t - t_m)}} \quad (2)$$

where, Δt and t_m represent the characteristic duration from 10% K to 90% K , and time to reach 50% K respectively. In this study, Δt , and t_m are both assumed to be 15 years, while K will be assumed to different vales under various scenarios. The profile shape can be changed by adjusting Δt and t_m . Smaller Δt and t_m will generate a faster H_2 demand growth.

Power, industry, and domestic sectors

The distribution of H_2 in 2050 to meet power, industry, and domestic energy demands under low, medium, and high scenarios is shown in Fig. 2. Currently, around 96% of electricity is generated from natural gas (NG) in Singapore, while solar PV contributes less than 3%. The rest is from petroleum products, coal, and peat. We assume that 5 vol%, 10 vol% and 15 vol% H_2 will be blended with natural gas for power plants in the early years (2021–2025) for the three scenarios,

respectively. Since the energy density of natural gas by volume is around 3.4 times of that of hydrogen, the initial H_2 share (by energy content) in the power sector will be 1.5%, 3.0%, and 4.5% respectively. Then, the H_2 share will gradually increase to 40%, 50%, and 60% by 2050, assuming gradual decommissioning and replacement of existing natural gas turbines by new 100% hydrogen gas turbines. And 20%, 30%, and 40% of power sector energy demand will be met by solar electricity [39]. The remaining will be produced from natural gas. It is also assumed It is also assumed petroleum products, coal, and peat will be completely ruled out by 2050.

In 2019, the average power generation efficiency was 50.9% for natural gas (NG), 22.0% for petroleum products, 20.2% for coal and peat, and 18.9% for others (mostly solar PV), as reported by the Singapore Energy Market Authority [40,41]. We assume the average efficiency will increase by 0.2% each year, e.g. 50.9% via NG in 2020 and 57.1% via NG in 2050. We also assume the power generation efficiency from NG and H_2 is the same. Details of the assumptions can be found in Section 2 of the Supporting Information.

Daily power demand in Singapore from 2021 to 2050 is projected to increase 11.4 TJ per year, based on the historical trends from 2005 to 2019 (Section 1 of the Supporting Information). With the power demand, H_2 share, and power generation efficiency, the H_2 demand profile for the power sector is generated, as shown in Fig. 3. H_2 is assumed in this model to replace natural gas as the fuel for the industry and domestic gas sectors. The H_2 demand is projected based on the industrial natural gas consumption and town gas consumption

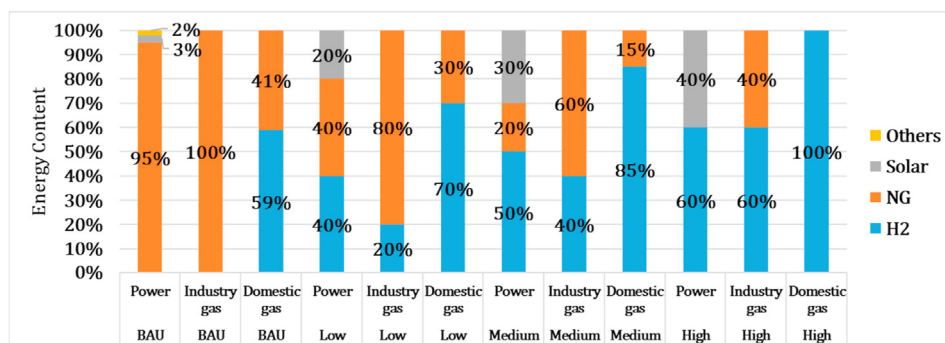


Fig. 2 – Share of H_2 , natural gas (NG), and solar for power, industry gas, and domestic gas sectors in 2050 under low, medium, and high scenarios.

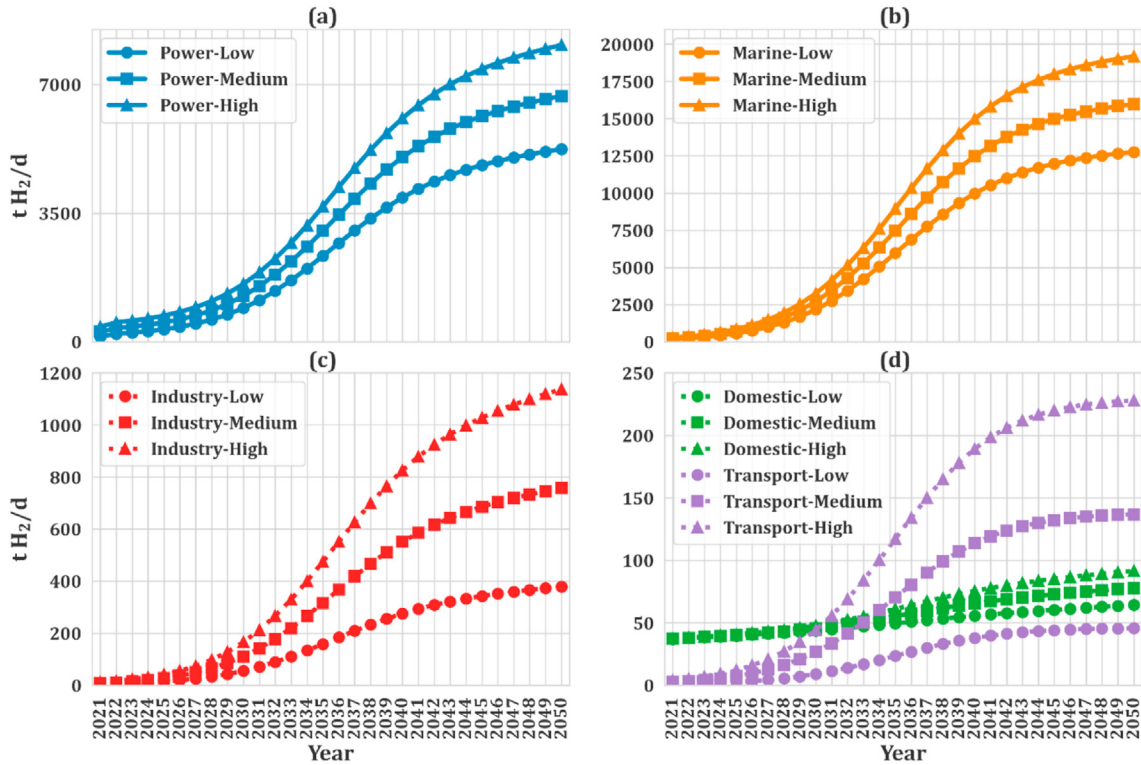


Fig. 3 – H₂ demand as tonne H₂ per day (t H₂/d) for power (a), marine (b), industry (c), domestic (d), and transport (d) sectors (2nd y-axis) from 2021 to 2050 under low, medium, and high scenarios. H₂ demand for the domestic sector is much smaller than others. Thus, three lines practically overlap with each other.

from 2013 to 2019 (Section 2 of the Supporting Information). Similarly, H₂ demand is also estimated for the low, medium, and high scenarios, which is also shown in Fig. 3. As mentioned before, the logistic substitution model is adopted to generate the H₂ share profiles.

Road transport sector

For the road transport sector, vehicles are classified into five types: cars (private cars), taxis, buses, trucks (goods vehicles), and motorcycles. Trucks can be further classified into three categories: light-duty trucks (≤ 3500 kg), medium-duty trucks (3501–11,000 kg), and heavy-duty trucks ($\geq 11,000$ kg) according to their maximum laden weight. The vehicle population [42] by fuel type is shown in Table 1. Fuel economies and average annual mileages of different vehicles can be found in Section 2 of the Supporting Information.

From 2009 to 2020 (11 years), the private car population increased by 10%, and the bus population increased by 21%. Thus, we assume that the car and bus population increase linearly at an annual rate of 0.9% (10%/11) and 1.9% (21%/11) respectively from 2020, as shown in Eq. (3) and Eq. (4).

$$Car_t = Car_{2020} \times (1 + 0.9\% \times (t - 2020)) \quad (3)$$

$$Bus_t = Bus_{2020} \times (1 + 1.9\% \times (t - 2020)) \quad (4)$$

From 2009 to 2020, the truck and motorcycle populations decreased slightly, while taxi population decreased by 37%. Since the taxi population is relatively small, a decreasing

scenario will make the population unreasonably small in 2050. Therefore, it is assumed the taxi population remains stable from 2021 to 2050. Similarly, we assume the truck and motorcycle populations also remain unchanged. The historical vehicle population from 2009 to 2020 can also be found in Section 2 of the Supporting Information.

Using the populations for various vehicle types from 2021 to 2050, we generate three scenarios with different battery electric vehicle (BEV), plug-in hybrid electric vehicle (PHEV), and H₂ fuel cell electric vehicle (FCEV) market penetrations [37,43]. Singapore government has proposed a BEV roadmap for road transport. It seems BEVs will dominate Singapore's road transport sector. However, for heavy-duty vehicles, there are still some challenges for BEVs. Thus, for the scenarios in this study, we assume H₂ FCEVs are adopted in the bus and truck fleets but not in cars, taxis, and motorcycles. All fleets see growth in BEVs and PHEVs. Correspondingly, the population of internal combustion engine vehicles (ICVs) declines and the fuel economy of those ICVs increase over time. We assume the average fuel economy (MPG, mile per gallon) of each vehicle fleet has an annual increasing rate of 1% compared to the current fuel economy, as follows:

$$MPG_t = MPG_{2020} \times (1 + 1.0\% \times (t - 2020)) \quad (5)$$

The total fuel consumption and power demand under three scenarios can be obtained using fuel economy and average annual mileage of different vehicles. The electricity demand for all BEVs and PHEVs of the high scenario in 2050 will reach 36.7 TJ/day, which is around 4% of the overall

electricity demand in Singapore. Note that, the electricity demand for the transport sector is also considered when projecting the H_2 demand for the power sector, as described in Section 2 of the Supporting Information.

Marine transport sector

International

Total sales of international marine bunker fuels in Singapore was 49.8 million tonnes in 2020 [44], equivalent to 2 million TJ/year. We projected marine fuel demand from 2021 to 2050 based on bunker fuel sales data from 2011 to 2020 (see Section 2 of the Supporting Information). Currently, lower sulfur fuel oil (LSFO) comprises 69% of the total bunker fuel sales, followed by marine fuel oil (MFO) with 21% share. We assume the hydrogen penetration into the marine fuel sector will be 20%, 25%, and 30% in 2050 for the low, medium, and high scenarios, respectively. This could come in the form of liquid ammonia (LNH3) or liquid hydrogen (LH2). For simplicity, we assume the market is split 50%:50% between these two fuels. The sum of liquid hydrogen and liquid ammonia demands, represented as H_2 equivalent, is also shown in Fig. 3. The H_2 demand for marine sector is the highest among all sectors, even with a relatively lower penetration compared to other sectors.

Domestic

For domestic navigation (tug boats, ferries, etc.), the total energy demand (gas/diesel) was 3224 TJ/year in 2020 [45], which is only 0.16% of the international marine bunker fuel demand. Decarbonization of domestic navigation should be easier compared to international marine bunkers, since there are less disruptive factors and it can be driven by local government policies. Therefore, we assume higher penetration levels under the low, medium, and high scenarios as 30%, 50%, and 70% in 2050, compared to international marine bunkers. Again, we assume the share of liquid hydrogen and liquid ammonia will be 50%–50%.

Case study on producing H_2 locally in Singapore

In this section, the possibility of local producing H_2 locally via SMR with CCS is explored for Singapore under low, medium, and high demand scenarios described above, namely Case L-Low, Case L-Medium, and Case L-High. The H_2 costs and CO_2 emission reduction potentials are evaluated and compared with the current energy sources. All the assumptions are given in Section 3 of the Supporting Information. Input data and numerical results can also be found in a spreadsheet of the Supporting Information.

Methodology

The optimization model is developed based on Fig. 1, where the total annual cost is minimized. The model is formulated and solved by the General Algebraic Modeling System (GAMS). The model inputs to carry out the case studies are provided using our in-house tool, Hydrogen Economy Assessment Resource Tool (HEART) [46,47].

The procedures to implement the case study include four steps. In the first step, basic information, such as electricity

and energy prices, is collected. A projection for the entire period of analysis is also generated for these parameters. In the second step, H_2 production technologies are selected. In this case, only SMRCCS is considered. Besides, H_2 demand projection is generated for different scenarios. Then, H_2 transport supply chains are constructed, considering various options for delivering H_2 from production sites to different end users. In the final step, the optimization model is carried out. The optimization outputs from GAMS are further analyzed to obtain the levelized cost of H_2 and emissions of H_2 . A flowchart can be found in Section 5 of the Supporting Information and details of the procedure can be found in a previous publication [46]. The results based on minimizing total cost over the 2020–2050 period for each of the three scenarios are presented below.

H_2 cost

From a consumer perspective, the high cost of hydrogen relative to incumbent fuels is a critical barrier to its adoption. The projected cost of H_2 into the sectors of interest are shown in Fig. 4: the cost of 20-bar H_2 (CH20) for power, industry, and domestic sectors and the costs of liquid ammonia (LNH3) and liquid hydrogen (LH2) for marine applications, and the costs of 700-bar compressed H_2 (CH700) and 350-bar compressed H_2 (CH350) for road transport. Note that CH700 and CH350 costs are at the refueling station dispenser, while CH20, LNH3, and LH2 costs are at the outlet of the distribution pipeline.

The H_2 cost is affected by many factors: SMR plant capacity, carbon tax, CCS cost, etc. The H_2 cost profiles of Fig. 4(a) for all three cases have overall declining trends until 2032

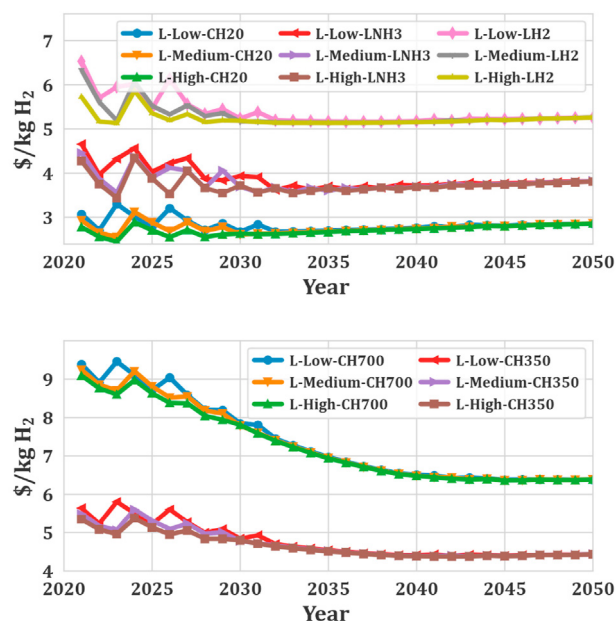


Fig. 4 – (a) 20-bar H_2 (CH20), liquid ammonia (LNH3) cost (represented as H_2 equivalent), and liquid hydrogen (LH2) cost for Case L-Low, Case L-Medium, and Case L-High from 2021 to 2050. (b) 700-bar H_2 (CH700) cost and 350-bar H_2 (CH350) cost for Case L-Low, Case L-Medium, and Case L-High from 2021 to 2050.

changing to an overall mildly increasing trends. The decline is due to the decreasing production cost from the economies of scale as demand increases and larger plants are built. However, for each newly built plant there is a lag time until its capacity is fully utilized. The oversized plants lead to a higher annual capital cost per unit H₂ produced which decreases for the next one/two years due to (slowly) increasing demand until the next (oversized) plant is built when the cost goes up again. This results in the periodicity (within the overall declining trends observed in the cost profiles until around 2032. The cost reduction from economies of scale is less important when the demand is high at the beginning. As shown in Fig. 4(a), the slopes of cost curves for Cases L-Low are relatively small.

The reason for the mildly increasing trends beyond 2032 is that the impact of economies of scale gradually vanishes as demand climbs while the impact of increasing carbon tax becomes significant. We count the emissions from both HSC and LNG supply chains. The total carbon intensity decreases over time because emissions from the NG supply chain and grid electricity decrease over time, which results in the mildly increasing trend. The carbon intensity decreases from 3.5 to 2.4 kg CO₂/kg H₂ for CH20, from 4.5 to 2.9 kg CO₂/kg H₂ for LNH3, and from 7.0 to 4.0 kg CO₂/kg H₂ for LH2.

The cost difference between CH20 and LNH3 is around 1 \$/kg of H₂ equivalent. If we assume that cost of pipeline transport of CH20 is negligible, then this difference is primarily the cost of the liquid ammonia (synthesis, storage, and pumping). The cost of LH2 is about 1.4 \$/kg higher than LNH3 cost due to the higher energy and cost for liquefaction. CH700 and CH350 costs show an overall decreasing trend. There are two main reasons: increasing utilization factor of refueling stations and decreasing capital cost of refueling stations. We assume the utilization factor of refueling station will increase from 60% to 90% as more H₂ FCEVs are adopted. The carbon intensity of CH700 and CH350 decrease from 5.5 to 3.5 and 4.7 to 3.1 kg CO₂/kg H₂ respectively.

Roadmap for production infrastructure and CCS

In this section, we determine when and what size of hydrogen infrastructure should be built to meet the projected demand. The total H₂ production capacity (metric ton per day, or TPD), average operating capacity factor (%), and the capacity of new plants (TPD) from 2021 to 2050 for our three scenarios are shown in Fig. 5. SMRCCS plants with a total nominal capacity of 32,000 TPD in 2050 (Case L-High) are required for the complete transition to H₂ fuel in the transport (marine and road), power, industry, and domestic sectors. Demand is dominated by the marine transport sector (67%) and power sector (28%) respectively. An 341-TPD SMRCCS plant will need about 80,000 m² (20 acres) land area [48], while an 800-TPD ATRCCS (Autothermal Reforming with CCS) plant will need about 55,000 m² land area including access roads [9]. For 32,000 TPD capacity, it will need about 7.5 and 2.2 km² respectively.

To meet the yearly demand growth, one can build new production plants or increase the operating capacity of existing plants. The optimization indicates that one should build oversized plants at suitable intervals and increase the operating capacity progressively in the interim period to meet the

H₂ demand, shown by the periodicity in the early years in the average operating capacity factor curves in Fig. 5. The frequency of building new plants is high in all three scenarios in the early period when the demand is still low but increasing rapidly. The underutilization period of each plant is generally short, and the average operating capacity in most of the years during the entire 30-year period is greater than 80%. This behavior emerges because the penalty from oversized plants during the underutilization period is offset by CAPEX saving from an economy of scale when building bigger plants. Hence, it is affected by the scaling factor of CAPEX, as well as the demand profile, which led to different frequency of building new plants for the three cases. For example, the H₂ demand is not only small but also increasing slowly in the first ten years for Case L-Low. The optimized result shows four small-scale oversized plants are built. During the period from 2032 to 2041, new plants are added every year to meet the increasing demand. The capacity is even higher than the maximum capacity for a single-train SMR plant, which is around 600 TPD [9]. Multiple trains are required for larger capacities.

To reduce the carbon intensity of H₂, most of the CO₂ will need to be captured from the SMR plants, resulting in what is known as blue H₂. The amount of CO₂ captured can reach more than 250,000 TPD in 2050 in Case L-High, as shown in the Section 3 of the Supporting Information. Even for Case L-Low with less than 50% transition to H₂ in 2050, more than 150,000 TPD of CO₂ is captured in 2050. This presents a challenge, since Singapore has no local sequestration sites. In this study, we assume that CO₂ is stored in the Malay basin in the Gulf of Thailand which is about 700 km away from Singapore [49]. Both onshore and offshore CO₂ deliveries to the sequestration sites are possible. The total cost for delivering CO₂ by pipelines and sequestering is estimated to be 48.4 \$/t (±20%). The CO₂ capture and compression cost vary with capacity. For a 500 t/d SMRCCS plant, the capture and compression cost is around 52.0 \$/t (±20%). The total CCS cost for the SMR plants in Singapore is about 100 \$/t CO₂. Details about CO₂ delivery and sequestration costs are given in Section 1.6 of the Supporting Information. The large-scale CO₂ capture and sequestration burden will be a critical barrier to NG-based H₂ production in Singapore, particularly until there is more certainty around its costs.

Discussion

We now examine the implications of H₂ adoption on the economics and emissions of Singapore's demand sectors.

Power sector

LCOE

In the power sector, hydrogen will displace NG gradually as a fuel via combined cycle gas turbine (CCGT). The alternative is to use NG for power generation via CCGT with CCS. We use the levelized cost of electricity (LCOE) for these pathways to compare their economics. The capital costs of CCGT plants with and without CCS are obtained from IEA [50] and U.S. Energy Information Administration (EIA) [51] reports. Details are in Section A of the Supporting Information.

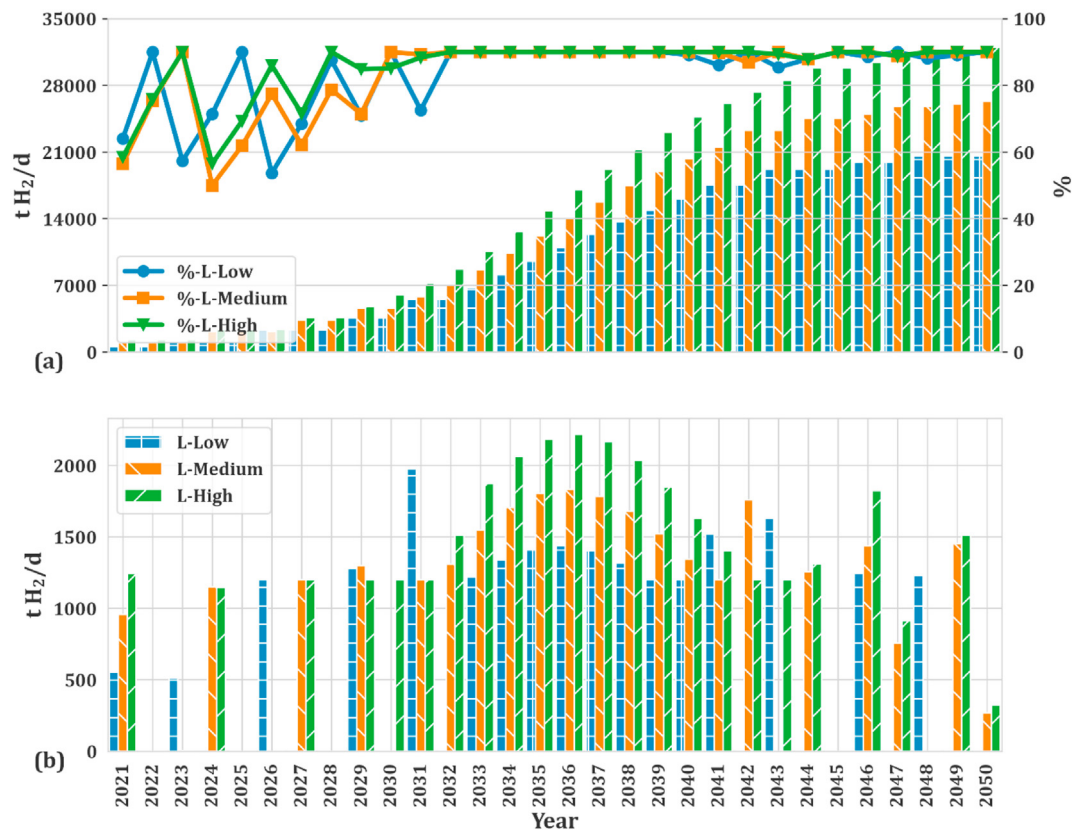


Fig. 5 – (a) Total H₂ production capacity (left y-axis, t H₂/d) and average operating capacity factor (right y-axis, %) and (b) the capacity of new plant (t H₂/d) from 2021 to 2050.

LCOE via H₂ under three scenarios and via NG with and without CCS are compared in Fig. 6. LCOE via H₂ follows the pattern of H₂ cost, while LCOE via NG CCGT without CCS increases rapidly. This is because a penalty for CO₂ emissions is applied, which increases over time. The slightly increase in LCOE via NG CCGT with CCS results from the penalty paid for 10% uncaptured CO₂ in the CCC units (90% of CO₂ is captured) and CO₂ emissions from the supply chain of PNG and/or LNG. LCOE via H₂ is around 50\$/MWh higher than the one via NG CCGT with CCS. CO₂ capture from SMR (~20%) is less energy-intensive and less expensive than from NG CCGT (~3% CO₂). The energy penalty for CO₂ capture and compression (CCC) via SMRCCS and CCGTCCS are 0.50 kWh NG [52] and 0.68 kWh NG

[50] per kg CO₂. However, the overall efficiency from NG to H₂ and then electricity is still lower than from NG to electricity (see Section A of the Supporting Information). Since the capital cost of CO₂ capture via CCGTCCS is only 1.3 times of the capital cost of CO₂ capture for SMRCCS per kg CO₂, the capital cost saving from CO₂ capture via SMRCCS does not offset the impact of this higher energy penalty. As a sensitivity analysis, we show a case where the capital cost of CO₂ capture is 3 times higher for CCGTCCS than SMRCCS (indicated by CCGTCCS') in Fig. 6. Such high CO₂ capture cost is required to bring LCOE parity between CCGTCCS and SMRCCS.

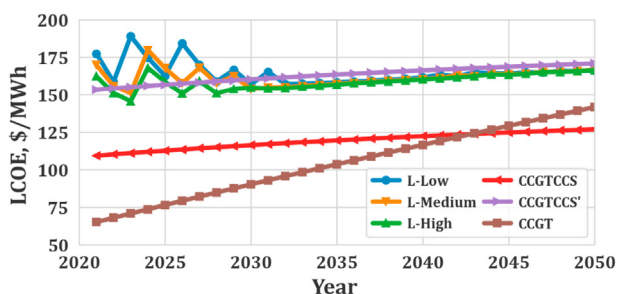


Fig. 6 – LCOE comparisons of H₂ power plants (L-Low, L-Medium, L-High) with natural gas power plant (CCGT and CCGTCCS, CCGTCCS2). CCGTCCS' assumes three times CO₂ capture and compression cost of SMRCCS.

Carbon intensity, carbon avoidance, and carbon avoidance cost
Carbon intensity (tonne CO₂ per MWh) is the amount of CO₂ emitted per unit of power generated shown in the Section 3 of the Supporting Information for H₂ CCGT, NG CCGT, and NG CCGTCCS. Carbon avoidance is the percent CO₂ reduction via H₂ or NG CCGTCCS versus NG CCGT when generating the same amount of electricity, which we find to be 59%–69% for H₂ and 69–77% for NG CCGTCCS when the CO₂ capture rate is 90% for both SMR and CCGT. The carbon avoidance cost is the effective cost to avoid each tonne of CO₂ emissions by moving from unabated NG CCGT to either H₂ or NG CCGTCCS. This metric is compared in Fig. 7. It starts from 439, 410, and 380 \$/tonne CO₂ from low, medium, and high scenarios respectively, while only 148 \$/tonne CO₂ for NG CCGTCCS. As the LCOE for unabated NG CCGT increases significantly due to the increasing carbon tax, the carbon avoidance cost decreases to

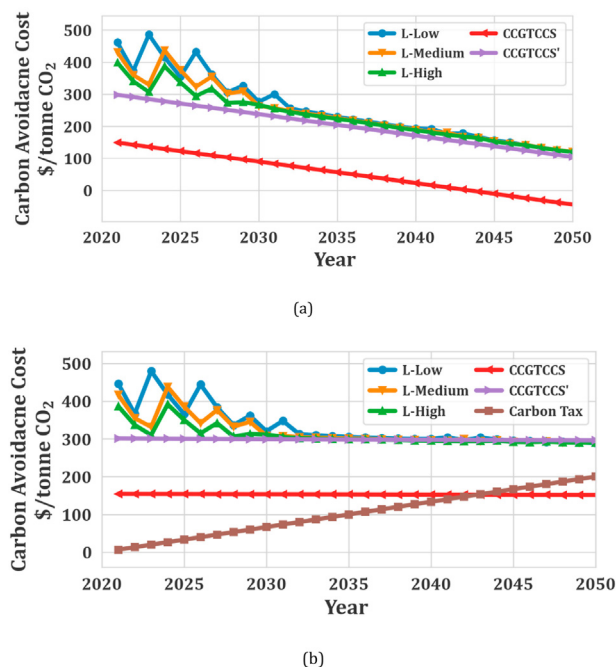


Fig. 7 – Carbon avoidance cost (\$/tonne CO₂) of H₂ CCGT and NG CCGTCCS, compared to NG CCGT. CCGTCCS' is calculated assuming the CO₂ capture and compression capital costs are 3X larger than for SMRCCS.

~90 \$/tonne CO₂ for H₂ and -48 \$/tonne CO₂ for NG CCGTCCS. The carbon avoidance cost calculated in the absence of a carbon tax is shown in Fig. 7(b). The carbon avoidance cost in Fig. 7(b) is equal to the sum of the one in Fig. 7(a) and carbon tax. The intersection of a carbon avoidance line and the carbon tax line indicates the level of carbon tax required to bring the two pathways into cost parity; e.g. blue H₂ in a CCGT and unabated NG CCGT. While this is observed for NG CCGTCCS at about 150 \$/tonne CO₂, it can be seen that an even higher carbon tax than we contemplate is required to bring H₂ into cost parity with unabated NG. In Section 3 of the Supporting Information, we also investigate the 'local carbon avoidance' cost, where only emissions in Singapore are considered (i.e. excluding NG supply chain emissions). The local carbon avoidance cost is found to be 12% and 4% lower than the carbon avoidance for the H₂ route and NG route, respectively.

Marine transport sector

As previously discussed, hydrogen could come in the form of liquid ammonia (LNH3) or liquid hydrogen (LH2) for marine transport sector. Based on LNH3 and LH2 costs in Fig. 4(a), the fuel cost for marine sector as \$/MWh energy content is shown in Fig. 10. Marine diesel oil (MDO, 450\$/tonne) and natural gas (NG, 25\$/MWh) as feedstock for LNH3 and LH2 are also shown in the same figure. The fuel costs of LNH3 and LH2 follow the trends in Fig. 4(a), while the fuel costs of NG and MDO increase due to carbon tax. Fig. 8(a) shows that LNH3 and LH2 produced from local blue H₂ is much more expensive than MDO, especially when the carbon tax is low.

Regarding the carbon avoidance cost, we assume the incumbent fuel is MDO. Fig. 8(b) shows it is higher than 500

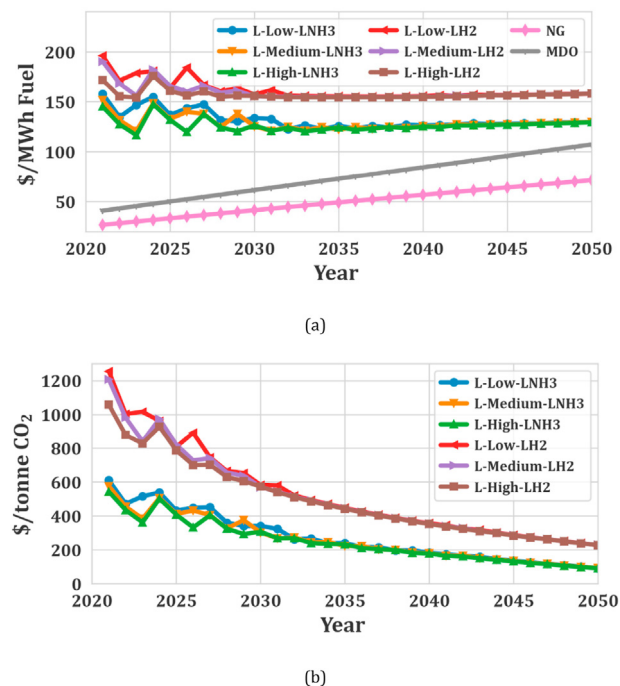


Fig. 8 – (a) Fuel Cost (\$/MWh) comparison among liquid ammonia (LNH3), natural gas (NG), liquid hydrogen (LH2), and marine diesel oil (MDO). (b) Carbon avoidance cost (\$/tonne CO₂) of liquid ammonia (LNH3) and liquid hydrogen (LH2), compared to marine diesel oil (MDO). Costs include penalty for CO₂ emissions during production and fuel combustion. Carbon avoidance cost calculated via removing carbon tax can be found in Section 3 of the Supporting Information.

and 250 \$/tonne CO₂ in the first ten years for LNH3 and LH2. The main reasons are: (1) high LNH3 and LH2 cost; (2) relatively high CO₂ emissions due to the electricity consumption during ammonia synthesis. The emissions per MWh fuel is 0.15–0.10 tonne CO₂ for LNH3, 0.22–0.12 tonne CO₂ for LH2, 0.25 tonne CO₂ for natural gas, and 0.34 tonne CO₂ for marine diesel oil. The carbon avoidance compared to MDO initially is only 56% and 36% for LNH3 and LH2, which increases to 71% and 65% respectively as the carbon footprint of LNH3 and LH2 decrease.

For domestic navigation, it is also possible to use CH700 and/or CH350 as fuel. However, the cost of CH700 and CH350 are both higher than the costs of LNH3 and LH2 according to Fig. 4(a) and (b).

Road transport sector

In this section, we evaluate the fuel cost for internal combustion vehicles (ICVs), battery electric vehicles (BEVs), and H₂ fuel cell electric vehicles (FCEVs) for passenger cars, buses, light-duty trucks, medium-duty trucks, heavy-duty trucks, and motorcycles on a \$/mile basis. This takes into account the fuel economy of various vehicles, which can be found in Section 2 of the Supporting Information. Note that the H₂ cost for road transport sector includes the cost of the refueling station. For ICVs, the fuel cost is based on the pre-tax gasoline

and diesel prices. For BEVs, the fuel cost is obtained from the LCOE in Fig. 6 and an additional charging station cost (15.2\$/MWh, see Section 1 of the Supporting Information) to it. Carbon tax is included in the LCOE. For FCEVs, the fuel cost is obtained from the cost of CH700 and CH350. Generally, light-duty vehicles (passenger car, light-duty truck, and motorcycle) use CH700, while heavy-duty vehicles (medium-duty truck, heavy-duty truck, and bus) use CH350. In Fig. 9, we show results for the high demand scenario where heavy-duty vehicles can use both CH700 and CH350.

Fig. 9 shows BEVs have the lowest fuel cost for all categories from 2021 to 2050. The main reason is a superior fuel economy (energy efficiency). As for FCEVs, the fuel cost is lower than ICVs for medium-duty truck, heavy-duty truck, and buses. It is higher than ICVs for light-duty vehicles (passenger cars, light-duty trucks, and motorcycles) before 2035. It is because CH700 is quite expensive at the early stage, due to high capital cost, high energy penalty, and low operating capacity factor of the refueling stations. As H_2 costs decrease and the penalty for CO_2 emissions increase, FCEVs become competitive with ICVs, but not BEVs. Despite the fact that fuel cost of BEVs and H_2 FCEVs are lower than that for ICVs, the

total cost could be higher, if the ownership cost is included. For example, the price of Toyota Camry (ICV), Telsa Model (BEV), and Toyota Mirai (H_2 FCEV) start from 25,000, 35,000, and 50,000 USD. By assuming a 10-year lifetime and 18,000 km per year travel distance, the ownership cost is about 0.42, 0.59, and 0.84 \$/mile. The fuel cost saving of passenger cars from BEVs (0.06–0.08 \$/mile) and FCEVs (0–0.03\$/mile) cannot offset the ownership cost penalty. But the ownership cost of BEVs and FCEVs are expected to decrease significantly in the future due to the higher demand and technological improvement.

The fuel cost saving of heavy-duty vehicles via BEVs or FCEVs is generally slightly higher, compared to light-duty vehicles. If CH350 is considered, hydrogen is already competitive with diesel, on a fuel cost per mile basis. Although it is still higher than for BEVs, the difference is very small. Since it is expected the H_2 FCEVs will have higher range than BEVs, CH350 based heavy-duty vehicles should draw more attention. However, if CH700 is adopted as heavy-duty vehicle fuel, it requires about 15 years to reach the breakeven cost.

Using the fuel cost (\$/mile) in Fig. 9 and carbon intensity (kg CO_2 /mile) in the Supporting Information, the carbon

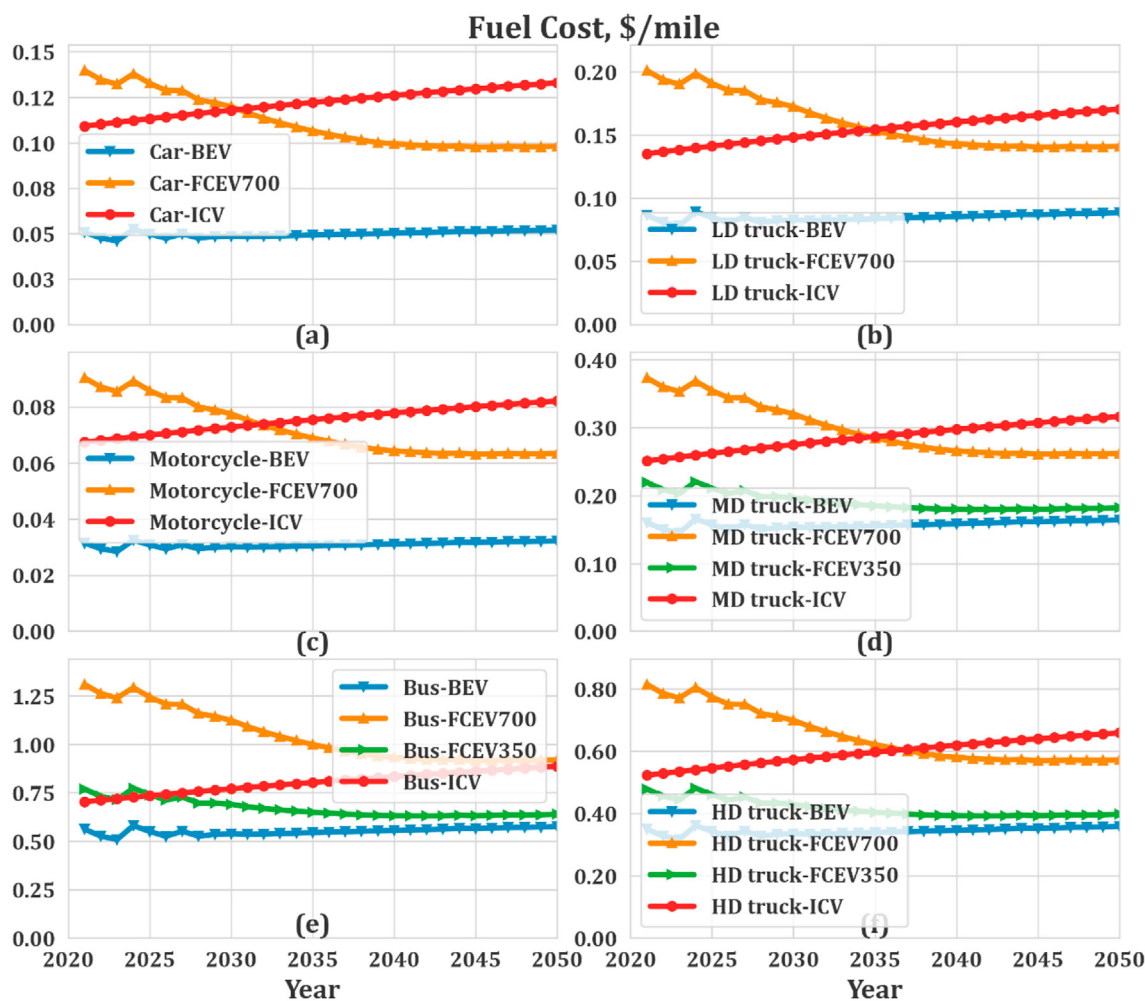


Fig. 9 – Fuel cost per mile (\$/mile) comparison among internal combustion vehicles (ICVs), battery electric vehicles (BEVs), and H_2 fuel cell electric vehicles (FCEVs). LD truck: light-duty truck, MD truck: medium-duty truck, HD truck: heavy-duty truck.

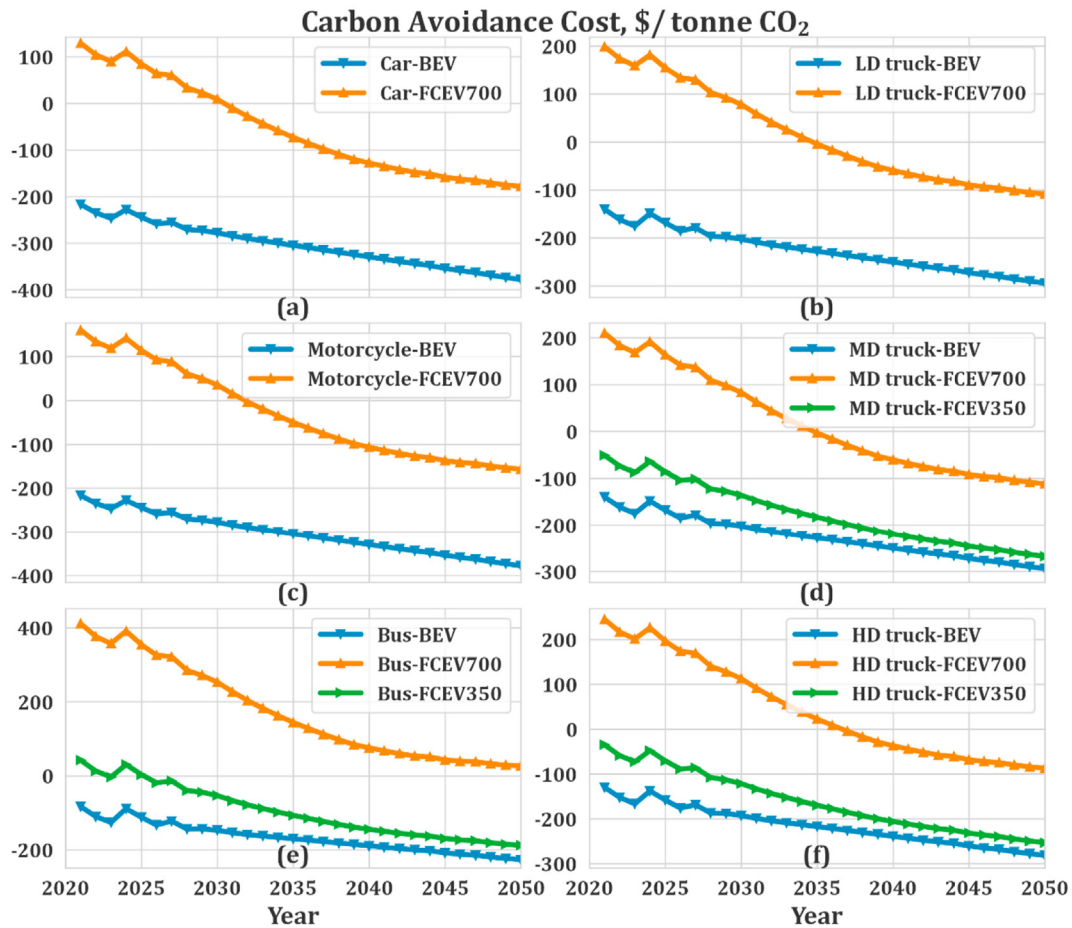


Fig. 10 – Carbon avoidance cost (\$/tonne CO₂) of battery electric vehicles (BEVs) and H₂ fuel cell electric vehicles (FCEVs), compared to internal combustion vehicles (ICVs). LD truck: light-duty truck, MD truck: medium-duty truck, HD truck: heavy-duty truck. Carbon avoidance cost calculated via removing carbon tax can be found in Section 3 of the Supporting Information.

Table 2 – Carbon intensity of BEs and H₂ FCEV fleets.

	Car	Bus	LD truck	MD truck	HD truck	Motorcycle
BEV	0.035~0.051	0.388~0.567	0.060~0.087	0.111~0.162	0.242~0.353	0.022~0.032
FCEV, 700 bar	0.054~0.084	0.502~0.788	0.068~0.104	0.143~0.225	0.313~0.490	0.035~0.054
FCEV, 350 bar	~	0.443~0.680	~	0.126~0.194	0.275~0.423	~

avoidance cost of BEVs and FCEVs compared to ICVs for six categories of vehicles are shown in Fig. 10. The reduction in carbon intensity for BECs is higher than H₂ FCEVs for each vehicle class fleet, as shown in Table 2. Since BEVs also have lower fuel cost, the carbon avoidance cost is smaller, which is negative as shown in Fig. 10. It indicates that there is no fuel cost penalty for decarbonization. For FCEVs, heavy-duty vehicles also have negative carbon avoidance cost, if CH350 is considered. Nonetheless, if CH700 is chosen, carbon avoidance cost starts at 100–400 \$/tonne CO₂ for different fleets and decrease to be negative, as CH700 becomes cheaper than gasoline or diesel.

Conclusions

In this paper, we investigated the local hydrogen supply chain roadmap for gradually shifting to a hydrogen economy in Singapore from 2021 to 2050. Only locally produced blue hydrogen was considered. Hydrogen's role in decarbonizing the transport, power, industrial, and domestic sectors in Singapore was explored. We evaluated the cost of H₂, the carbon emissions avoided, the carbon avoidance cost, and the corresponding supply chain under three different H₂ demand scenarios.

The key findings are

- (1) For a given increasing demand forecast, it is more economic to construct larger facilities than building new plants every year. However, the H₂ cost will be relatively high at the beginning due to low utilization of facilities (production, terminal, and refueling stations).
- (2) The local production capacity in Singapore could reach as high as 32,000 TPD to meet all H₂ demand, while a reliable CO₂ supply chain with the capacity up to 250,000 tonne CO₂ per day for remote sequestration of the captured CO₂ may be required.
- (3) Despite optimizing the H₂ supply chain to meet the projected H₂ demand profile, the cost of H₂ is not competitive with the other technologies we considered. The cost of CO₂ emissions avoided via hydrogen is higher than 100 \$/tonne CO₂ for the power and marine sectors. For the power sector, using SMRCCS to produce blue H₂ as a fuel to generate electricity is 30–50% more expensive than directly using NG CCGT with CCS.
- (4) For marine transport, LN₃ is less expensive than LH₂.
- (5) While BEVs constitute the dominant option for road transport, H₂ FCEVs should be considered for heavy-duty vehicles.

In order to overcome the barriers of local hydrogen production for Singapore's decarbonization, alternatives such as importing blue H₂ produced in countries with access to low-cost CO₂ storage should be considered. Besides, creating a hydrogen ecosystem mostly based on imported natural gas might have energy security and price volatility issues, the diversity of energy sources should also be considered.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2022.12.069>.

Abbreviations

ATRCCS autothermal Reforming with CCS
BEV battery electric vehicle
CAPEX Capital expenditure

CCGT combined cycle gas turbine
CCGTCCS combined cycle gas turbine with CO₂ capture and storage
CCS CO₂ capture and storage
CH₂₀ 20 bar H₂
CH₃₅₀ 700 bar H₂
CH₄₀ 40 bar H₂
CH₇₀ 700 bar H₂
FCEV fuel cell electric vehicle
GAMS General Algebraic Modeling System
HD truck heavy-duty truck
HEART Hydrogen Economy Assessment & Resource Tool
HSC hydrogen supply chain
ICV internal combustion vehicle
LCOE levelized cost of electricity
LCOH levelized cost of hydrogen
LD truck light-duty truck
LH₂ liquid hydrogen
LNG liquefied natural gas
LN₃ liquid ammonia
LSFO lower sulfur fuel oil
MDO Marine diesel oil
MD truck medium-duty truck
MFO marine fuel oil
MPG mile per gallon
MILP mixed-integer linear programming
MINLP mixed-integer non-linear programming
NG natural gas
PNG pipeline natural gas
PHEV plug-in hybrid electric vehicle
SMR steam methane reforming
SMRCCS steam methane reforming with CO₂ capture and storage
TC total cost
TCO₂ total CO₂ emissions
TPD tonne per day

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